

Projecte Intermodular – UT0C

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Link to Trello board: <https://trello.com/b/DAAAdSBPs/projecte-intermodular>

1. Initial Backlog (for Trello)

ID	Task Title	Description	Priority (MoSCoW)
BK-01	Initial Environment Setup	Create GitHub repository, configure Node.js and React project structure.	MUST
BK-02	Database Design	Define "Products" schema (title, artist, price, image, audio URL).	MUST
BK-03	Home Page	Implement a grid view displaying "Latest Releases".	MUST
BK-04	Product Detail Page	Show full album info, tracklist, and large cover art.	MUST
BK-05	Shopping Cart	Implement logic to add/remove items and calculate total price.	MUST
BK-06	Payment Gateway (Ethereum)	Basic integration with Wallet (e.g., MetaMask) to send funds.	MUST
BK-07	Audio Player	Component to listen to audio previews on the product page.	SHOULD

BK-08	Shipping Form	Input fields for the user to enter their physical delivery address.	MUST
BK-09	User Registration	Allow users to sign up and save their shipping details.	SHOULD
BK-10	Special Offers Section	A specific page filtering albums with discounted prices.	COULD
BK-11	Discogs API Integration	Automatically fetch tracklists and technical info using release ID.	COULD
BK-12	Admin Panel	Simple interface for the administrator to upload new products.	MUST

2. Sprint 1 Planning (January)

Sprint Goal: Create the base web application structure, configure the database, and enable product catalog visualization (Home and Detail pages).

User Stories:

1. As a user, I want to see a list of the latest releases so I know what's new in stock.
2. As a user, I want to click on an album to see its cover art and tracklist.
3. As an admin, I want the database connected so I can store product information.

Estimated Tasks (Sprint Backlog):

- Project configuration (React + Node.js) and Repository setup. (4h)
- MongoDB Database design and creation ("Albums" collection). (3h)
- Backend API development to fetch product list. (5h)
- "Home Page" layout implementation using CSS Grid/Flexbox. (6h)
- "Product Page" layout with dynamic components. (5h)
- **Total Estimated Time:** 23 hours of development.

3. Cost Table & Viability

Concept	Detail	Unit Cost	Total Cost
Development (Labor)	Estimate of 150 total project hours (Junior Dev).	20 €/hour	3,000 €
Web Domain	.com domain registration (annual).	15 €/year	15 €
Hosting (Server)	Basic VPS or PaaS (e.g., Heroku/Vercel).	10 €/month	120 € (1st year)
Software / Tools	GitHub (Free), Figma (Free), Trello (Free).	0 €	0 €
Initial Marketing	Social media campaign (optional).	200 €	200 €
INITIAL TOTAL			3,335 €

Economic Viability: The project has a low initial cost because it leverages open-source technologies (MERN Stack). The primary expense is labor. Annual maintenance is minimal (~135€/year), making the project highly viable if a regular sales volume is achieved. Using cryptocurrency eliminates traditional banking fees for maintaining POS terminals.

4. Final Reflection

This project combines a cultural passion (vinyl records) with emerging technology (decentralized payments). The current planning allows for a solid start in Sprint 1 by focusing on the visual catalog, while deferring the complexity of Blockchain payments to a later phase (Sprint 2 or 3). The use of the MERN stack will ensure a modern, fast User Interface (UI), which is essential for the shopping experience and audio previewing.